Replication Report: Financial-Network Unveiling

Course replication exercise

August 9, 2026

1 The Leontief unveiling operator

1.1 Economic object and notation

Mian, Straub, and Sufi (2025, Sections 2.1–2.2) ask who ultimately owns a financial claim when the observed holder is itself a financial intermediary. Let $\bar{M}_t$ denote the direct ownership matrix. Its entry $\bar{M}_t(i,j)$ is the share of sector i's liabilities held directly by sector j. Rows are therefore issuers or borrowers, columns are holders or lenders, and every row sums to one.

This normalization is row-specific. For example, the outgoing entries from a bank all divide by that bank's total liabilities, so they describe who funds the bank and sum to one. An incoming entry such as $\bar{M}_t(F,B) = 0.20$ instead says that the bank holds 20 percent of fund F's liabilities; its denominator is F's liabilities, not the bank's assets. Columns therefore need not sum to one. Intermediaries do own assets directly, but they are not terminal owners: their assets are financed by liabilities and equity that are themselves held by other sectors.

Order the m ultimate owners before the k intermediaries. After zeroing the owner rows, the matrix relevant for intermediation has block form

$$M_t = \begin{bmatrix} 0 & 0 \\ B_t & Q_t \end{bmatrix}, \tag{1}$$

where $B_t \in \mathbb{R}^{k\times m}$ contains intermediaries' direct owner shares and $Q_t \in \mathbb{R}^{k\times k}$ contains intermediary cross-holdings. The paper writes the sum over all ownership paths as

$$\bar{\Omega}_t = M_t + M_t^2 + M_t^3 + \cdots = M_t(I - M_t)^{-1}. \tag{2}$$

The lower-left block is the required ultimate-ownership matrix,

$$\Omega_t = B_t + Q_tB_t + Q_t^2B_t + \cdots = (I - Q_t)^{-1}B_t. \tag{3}$$

The maintained condition $\rho(Q_t) < 1$ ensures that intermediary cross-holdings eventually reach an ultimate owner. Economically, the network cannot contain a closed group of intermediaries that own 100 percent of one another without any owner endpoint.

1.2 Our implementation

The Python class in `Code/unveiling/network.py` validates the completed direct-ownership matrix once at construction. The central method then computes

$$(I - Q_t)\Omega_t = B_t \tag{4}$$

with `numpy.linalg.solve`. This is algebraically identical to Equation (3); it does not replace the paper's economic model or introduce a different allocation rule. Solving the linear system is preferable to explicitly forming $(I - Q_t)^{-1}$ because it states the fixed point directly and avoids an unnecessary matrix inversion.

The implementation checks the economically relevant invariants: rows of $\bar{M}_t$ and Ω_t sum to one, the spectral radius is below one, the closed-system net-worth identity sums to zero, and final primary-asset shares remain exhaustive. Unit tests also compare the block solve with both the paper's full expression $M_t(I - M_t)^{-1}$ and an 80-round path sum.

1.3 A complete network example

Consider households H, government G, a bank B, and a fund F:

$$\bar{M} = \begin{bmatrix} 1.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 & 0.0 \\ 0.2 & 0.1 & 0.0 & 0.7 \\ 0.6 & 0.2 & 0.2 & 0.0 \end{bmatrix}, \qquad B = \begin{bmatrix} 0.2 & 0.1 \\ 0.6 & 0.2 \end{bmatrix}, \qquad Q = \begin{bmatrix} 0.0 & 0.7 \\ 0.2 & 0.0 \end{bmatrix}. \tag{5}$$

The bank's liabilities are held 20 percent by households, 10 percent by government, and 70 percent by the fund. The fund is held 60 percent by households, 20 percent by government, and 20 percent by the bank. This reciprocal bank–fund link creates paths of arbitrary length even though the network has only four sectors.

Let ω_{BH} and ω_{FH} denote the bank's and fund's ultimate household shares. Reading the two rows of the network gives

$$\omega_{BH} = 0.20 + 0.70\omega_{FH}, \tag{6}$$
$$\omega_{FH} = 0.60 + 0.20\omega_{BH}. \tag{7}$$

Substituting the second equation into the first yields

$$\omega_{BH} = 0.62 + 0.14\omega_{BH} \quad \Longrightarrow \quad \omega_{BH} = \frac{0.62}{1 - 0.14} = 0.72093. \tag{8}$$

The same number can be read as direct plus indirect ownership:

$$\underbrace{20\%}_{\text{bank} \to \text{household}} + \underbrace{70\% \times 74.419\%}_{\text{bank} \to \text{fund} \to \text{ultimate household}} = 72.093\%. \tag{9}$$

The individual path contributions are

$$20.000\% + 42.000\% + 2.800\% + 5.880\% + 0.392\% + 0.823\% + \cdots = 72.093\%. \tag{10}$$

Thus the 72.093 percent is not an additional assumption: it is the sum of all direct and recursively intermediated paths encoded in the original matrix.

2 Data architecture: which source identifies which object

The datasets are complementary rather than interchangeable. The Financial Accounts identify the aggregate balance sheets and issuer–holder network; DINA identifies how the household sector is distributed across wealth groups and broad asset classes; NIPA supplies aggregate income, consumption, and saving identities; and the remaining surveys and price data either identify valuation terms or validate the baseline estimates. Figure 1 summarizes the three main empirical routes.

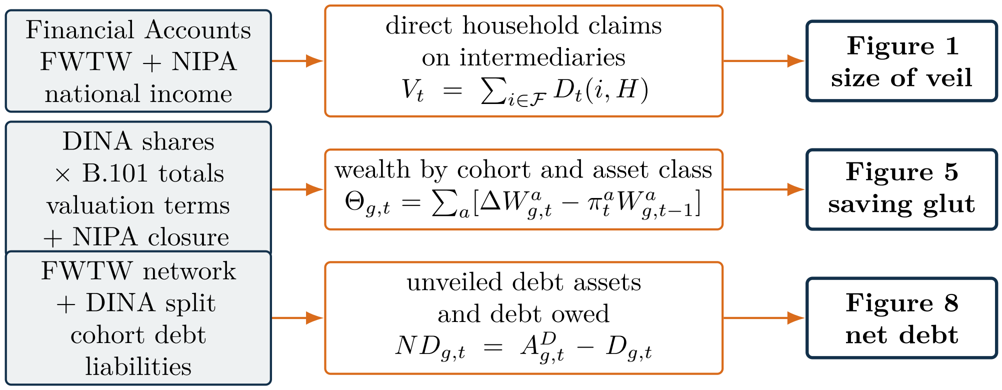

Figure 1: Dataset roles in the three selected July 2025 exhibits. DFA, the SCF panel, and income-minus-consumption estimates are validation layers, not substitutes for the baseline inputs.

2.1 The direct network is a Financial Accounts object

For each of the $\ell = 34$ instruments, let $X_{h,t}(i,j)$ be the dollar stock issued by sector i and held by sector j. Financial Accounts issuance and holding totals impose the margins

$$r_{iht} = \sum_j X_{h,t}(i,j), \qquad c_{jht} = \sum_i X_{h,t}(i,j). \tag{11}$$

Detailed Financial Accounts fields supply exact bilateral cells where they are available. If a margin contains exactly one unknown cell, that cell can indeed be recovered as a simple residual. With several unknown cells, however, the margins do not identify a unique matrix. The 2025 paper follows the Batty et al. procedure: known relationships and exclusion restrictions are retained, and the remaining cells are completed using proportionality and a balancing algorithm that respects known row and column totals. DINA does not enter this completion. After completing each $X_{h,t}$, the paper constructs

$$\bar{M}_t = \operatorname{diag}(L_t)^{-1} \sum_{h=1}^{34} X_{h,t}, \qquad \bar{M}_t \mathbf{1} = \mathbf{1}, \tag{12}$$

where $L_{i,t}$ is total liabilities issued by sector i. Thus the intuition that there are 23 instruments needs one dimensional correction: the paper has 23 intermediary sectors, four ultimate-owner sectors, and 34 instrument matrices.

2.2 DINA expands the household owner; it does not supply counterparties

DINA is a tax-record-based distributional microfile. The authors use it to estimate

$$\omega^a_{g,t} = \frac{\text{household wealth class } a \text{ held by cohort } g}{\text{aggregate household wealth class } a}, \qquad \sum_g \omega^a_{g,t} = 1. \tag{13}$$

For the saving calculation, these shares distribute the corresponding Financial Accounts B.101 total:

$$W_{g,t}^{a}=\omega_{g,t}^{a}W_{t}^{a}. \tag{14}$$

For distributional unveiling, Appendix Table A1 maps each Financial Accounts instrument h to a broad DINA asset class $a(h)$. Operationally, a direct household position can then be split into cohort columns as

$$X_{h,t}^{\text{aug}}(i,g)=X_{h,t}(i,H)\omega_{g,t}^{a(h)}. \tag{15}$$

The unveiling is repeated on the augmented matrix. Consequently, DINA is not a dataset of observed (wealth group, intermediary, instrument) counterparties. It supplies group shares for mapped household asset and liability classes. Combining those shares with Financial Accounts totals creates the distributional columns.

2.3 How the selected figures use the layers

Table 1: Main datasets and their economic roles

Source	**Observed object**	**Role in this project**
Financial Accounts / FWTW	Sector–instrument stocks; issuer–holder positions; B.101 household balance sheet	Builds $X_{h,t}$, $\bar{M}_t$, and Ω_t. It identifies Figure 1's aggregate amount behind the veil and the primary assets used in Figure 8.
DINA (`PSZusdina`)	Individual tax-record-based income and imputed wealth components, collapsed to cohort–asset shares	Baseline distribution of B.101 wealth for Figure 5; splits the household owner into wealth cohorts for Figure 8. It is not needed for aggregate Figure 1.
NIPA	Aggregate national income, disposable income, consumption, personal and corporate saving	Gives Figure 1's national-income denominator and forces distributional saving in Figure 5 to aggregate to national-account private saving.
JST and debt write-down inputs	House-price appreciation; write-down rates by debt type and group	Identify pure valuation terms π_t^a. Equity price inflation is the residual that closes aggregate wealth-based saving to NIPA.
DFA and SCF panel	Alternative wealth shares; 1983–1989 household wealth panel	Validate DINA wealth shares and repeated-cross-section saving estimates.
SCF consumption, CEX, and PSID	Survey expenditures, income, and household characteristics	Support the Appendix C.1 income-minus-consumption robustness exercise; CEX and PSID do not represent the richest households well enough to be the 2025 baseline.

Figure 1 therefore needs the Financial Accounts/FWTW network and NIPA only for the NI_t normalization; neither DINA nor consumption microdata are required. Figure 5 instead uses the wealth-based identity

$$\Theta_{g,t}=\sum_{a}\left[W_{g,t}^{a}-W_{g,t-1}^{a}-\pi_{t}^{a}W_{g,t-1}^{a}\right]. \tag{16}$$

Here π_t^a is pure asset-price inflation, not the total return. The JST repeat-sales house-price index identifies housing valuation, fixed-income price inflation is zero under the Financial Accounts convention, debt write-downs are treated as borrower valuation gains, and equity inflation is chosen residually so that $\sum_g \Theta_{g,t}$ matches total private saving in NIPA.

Income less consumption, $\Theta_{g,t} = Z_{g,t} - T_{g,t} + R_{g,t} - C_{g,t}$, is a separate robustness method in Appendix C.1. DINA supplies pre-tax income, taxes, and transfers; the authors use SCF-based consumption shares and CEX evidence to scale the observed consumption categories. The paper explicitly cautions that CEX and PSID do not measure consumption at the very top accurately. It is therefore incorrect to describe CEX or PSID as the baseline route to Figure 5.

Finally, Figure 8 uses the augmented unveiling result $A_t = P_t\Omega_t + \Lambda_t$. If $A^D_{g,t}$ is household debt held as an asset by cohort g and $D_{g,t}$ is debt owed by that cohort, then $ND_{g,t} = A^D_{g,t} - D_{g,t}$. The asset side requires the unveiled FWTW network; both the asset and liability sides require the household cohort mapping. NIPA enters only when the resulting stocks are normalized by national income.

3 Comparison with the authors' 2021 implementation

3.1 Version boundary

The available replication package is for the February 2021 paper, whereas the target article is the heavily revised July 2025 draft. The 2021 Stata code does not create the 2025 paper's 27×27 completed ownership matrix and does not evaluate Equation (2). Instead, `MSS2021Febreplicationkit/code/build_fa_unveiling_data.do` allocates household debt, government debt, and mortgage debt through separate, instrument-specific sequences.

For household debt, round 1 identifies the direct holders and checks their sum with `test1` (lines 131 and 166–171). Subsequent round markers occur at lines 724, 1066, 1259, 1740, 2138, and 2316. At each stage the script passes the debt embedded in selected intermediary assets to the holders of those intermediaries' liabilities. The final stage constructs owner and residual categories at lines 2385–2388 and saves `Yunveilhhd.dta` at line 2405. Government and mortgage debt repeat the procedure in separate code blocks.

Table 2: The two implementations solve related, but not identical, empirical problems.

	2025 paper and our Python	**February 2021 Stata package**
Representation	One completed direct-ownership matrix $\bar{M}_t$	Instrument- and sector-specific dollar variables
Recursion	The same B_t, Q_t operate at every path length	Different instruments and sector rules are opened at named stages
Stopping rule	Exact infinite sum when $\rho(Q_t) < 1$	Seven economically ordered stages plus residual buckets
Data treatment	Completed FWTW cells are an upstream input	Proportionality factors, net issuance, and special cases are coded inline
Conservation	Owner shares sum to one	`test1`–`test7` preserve the relevant debt stock
Output	Share matrices Ω_t and A_t	Dollar totals assigned to household, government, foreign, and residual owners

3.2 What can be proved to be the same

Suppose an explicit round-by-round algorithm is applied to the same fixed B_t, Q_t used by the matrix method. After R rounds it produces

$$\Omega_t^{(R)} = \sum_{r=0}^{R-1} Q_t^r B_t. \tag{17}$$

If $\rho(Q_t) < 1$, the finite sequence converges to the Leontief solution:

$$\lim_{R\to\infty} \Omega_t^{(R)} = (I - Q_t)^{-1} B_t = \Omega_t. \tag{18}$$

The remaining truncation error has the exact expression

$$\Omega_t - \Omega_t^{(R)} = Q_t^R \Omega_t. \tag{19}$$

This establishes algebraic equivalence between a stationary round algorithm and our linear solve. In the worked example, seven terms have a maximum error of 0.00143 in ownership shares, or 0.143 percentage points. The 80-term test agrees with the closed form to an absolute tolerance of 10^{-12}.

This numerical exercise must not be confused with running the actual seven rounds in the authors' Stata file. Their stages are not seven powers of one fixed Q_t: they unveil different instruments, alter sector treatments, use net-issuance adjustments, and retain explicitly defined residuals. Equation (19) therefore proves the common mathematical principle, not equality of the two historical empirical pipelines.

3.3 Evidentiary conclusion

The present evidence supports three deliberately bounded conclusions.

1. **Exact algebraic consistency with the 2025 paper.** Our block solve is the lower-left block of the paper's Equation (1), and the unit tests establish numerical equality on a common matrix.

2. **Conceptual consistency with the 2021 package.** Both methods recursively attribute an intermediary's embedded assets according to the ownership of its liabilities, while preserving the total amount being allocated.

3. **No completed empirical equality test across versions.** The 2021 package and 2025 exposition differ in inputs, instruments, sector definitions, residual treatment, and sample period. The completed 2025 annual matrices are not included in the old package.

It would therefore be incorrect at this stage to claim either that the two actual implementations lead to the same empirical series or that a difference shows an error in the paper. Nor is the Python method unconditionally "better." It is shorter, numerically exact for the stationary operator, and closer to the 2025 exposition. The Stata implementation is more explicit about the instrument-specific empirical adjustments used in the earlier paper. Those are different strengths attached to different versions of the research design.

3.4 Protocol for the eventual numerical comparison

A defensible empirical comparison requires a common-input experiment:

1. construct the same completed issuer–holder dollar matrices and sector aggregation for both algorithms;

2. derive B_t and Q_t and run the Python Leontief solve;
3. reproduce the authors' relevant allocation rules in an isolated copy of the Stata pipeline, preserving its residual definitions;
4. aggregate both outputs to identical owner, asset, date, and unit definitions;
5. compare cell-level and aggregate differences using a predeclared maximum absolute share tolerance and verify total-debt conservation; and
6. attribute any remaining discrepancy to truncation, residual treatment, missing-cell completion, net issuance, or a genuine coding difference.

Only after this protocol can the report make a supported statement about numerical equality. A claim that the paper is partially wrong would require a reproducible violation of its stated equations or accounting identities on the authors' own intended inputs, not merely a difference from the older package.

4 Reconstruction of Figure 1

4.1 What the figure measures

Figure 1 measures the size of the balance-sheet layer that must be unveiled. Let $D_t(i, H)$ be the dollar liabilities issued by sector i and held by the household sector, after summing the FWTW cells over instruments. If $\mathcal{F}$ is the set of financial-intermediary issuers, define

$$V_t = \sum_{i \in \mathcal{F}} D_t(i, H). \tag{20}$$

The household's claim in Equation (20) is direct with respect to the intermediary but indirect with respect to the primary assets that the intermediary finances. This is why it is the amount that "needs to be unveiled." It is not the elementwise difference between Ω_t and $\bar{M}_t$, which have different dimensions and economic meanings. The paper plots

$$\frac{V_t}{\sum_{i \neq ROW} D_t(i, H)} \qquad \text{and} \qquad \frac{V_t}{NI_t}, \tag{21}$$

where the first denominator is household holdings of U.S. financial assets and NI_t is annual national income. Both balance-sheet amounts are Q4 stocks in current dollars; national income is a current-dollar annual flow.

4.2 Three evidence layers

The empirical exercise must distinguish three questions. First, the February 2021 package can be run on its own final files to benchmark the older paper, but that package has no predecessor of the revised Figure 1. Second, we can reconstruct the Figure 1 numerator from the authors' supplied source vintage. Third, the June 2026 public FWTW release provides a new-data robustness exercise. The third layer cannot substitute for the second.

The 2025 article uses $k = 23$ intermediary sectors and $\ell = 34$ financial instruments. These numbers must not be conflated: there are 23 sectors, not 23 instruments. The February 2021 package explicitly says that its prepared `LQpanel_2019Q1.dta` omits the underlying FWTW text files. It therefore does not contain the completed 34 instrument matrices used by the July 2025 paper. The link labelled "replication kit" in the supplied 2025 PDF also resolves to the placeholder `NEW LINK.pdf`, so it does not supply the missing vintage.

4.3 Authors-data reconstruction from the February 2021 snapshot

We start from the following prepared inputs rather than rerunning every upstream import and merge:

- `LQpanel_2019Q1.dta`: the authors' quarterly wide Financial Accounts panel; we select Q4 stocks and the required household instruments;
- `Yunveilhhd.dta`: only the authors' already-computed allocation of aggregate corporate bonds to ABS issuers, finance companies, mortgage REITs, and depository institutions;
- `msf.dta`: the supplied licensed CRSP monthly security file, from which we construct the corporate-equity extension; and
- `YinequalityNIPAanalysis.dta`: the authors' annual national income series, which fixes the common-input sample at 1963–2016.

This skips mechanical source ingestion, table cleaning, and merges that the authors have already performed. It does *not* skip the economically central aggregation. Our code constructs

$$\begin{aligned} V_t^A = C_t + T_t + MMF_t + GSE_t + MF_t + LI_t + (PE_t - UPE_t) \\ + FB_t + FE_t, \end{aligned} \tag{22}$$

where the terms denote checkable deposits and currency, time and saving deposits, money-market-fund shares, GSE securities, mutual-fund shares, life insurance reserves, funded pension entitlements, financial-intermediary bonds, and listed financial-intermediary equity. These are the nine nonzero household-intermediary claim groups recoverable from the old snapshot. The remaining paper instruments either represent direct primary claims, have no household position in this vintage, or require the missing bilateral FWTW cells. Every component in Equation (22) remains in the generated annual data, and the code verifies that their sum equals V_t^A.

For the equity extension, we retain December CRSP observations, reproduce the authors' exclusion of foreign-incorporated share codes, and classify SIC 6000–6399 and 6700–6799 as listed financial intermediaries. Prices times shares outstanding are converted to millions of dollars. We then apply the old paper's proportionality logic,

$$FE_t = \frac{\text{household corporate-equity assets}_t}{\text{all corporate-equity assets}_t} \times \text{listed financial market capitalization}_t. \tag{23}$$

The implementation is `Code/unveiling/figure1_authors.py`; the thin entry point is `Code/scripts/build_figure1_authors_data.py`. Four tests check the component identity, unit conversion, denominator exclusions, and annual merge contract.

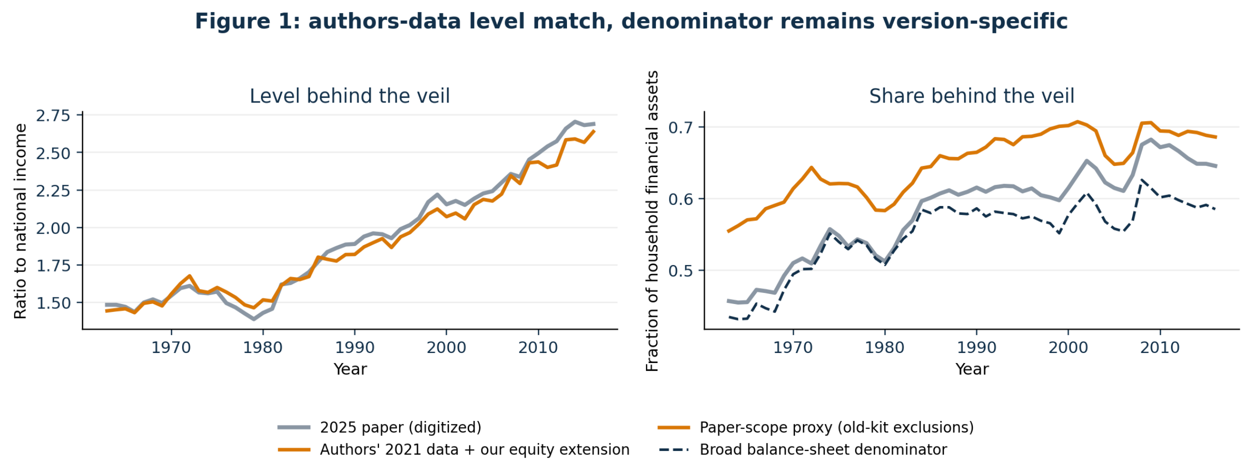

Figure 2: Figure 1 benchmark and reconstruction from the authors' February 2021 source vintage. The left panel is directly comparable. The two right-panel denominators diagnose the missing 2025 FWTW boundary.

Table 3: Quantitative authors-data comparison, 1963–2016

Statistic	**2025 figure, digitized**	**Our reconstruction**
Level in 1963	148.5%	144.4%
Level in 2016	269.0%	263.9%
Annual level MAE	5.3 pp	
Annual level correlation	0.995	
Share MAE, broad balance-sheet denominator	3.2 pp	
Share MAE, old-kit paper-scope proxy	6.6 pp	

4.4 The level is closely reproduced; the share denominator is not

We digitize the two colored lines from the raster embedded on physical PDF page 11. The digitization is a generated, approximate benchmark, independent of the reconstruction. A few pixels of line thickness imply approximately two percentage points of national income on the left axis and less than one percentage point on the right axis.

The left-axis result is strong: the reconstructed level has an annual correlation of 0.995 with the plotted line and an MAE of 5.3 pp of national income. It starts at 144.4% rather than 148.5% and reaches 263.9% rather than 269.0%. Thus the authors' older source data plus the explicit equity extension recover almost all of the plotted stock behind the veil.

The right-axis result is less definitive because its exact denominator needs the missing completed issuer–holder matrix. A broad funded household balance-sheet denominator is closer to the plotted series (MAE 3.2 pp) than the old kit's available implementation of the paper's stated nonfinancial/foreign/residual exclusions (MAE 6.6 pp). This ordering is evidence of a versioned measurement boundary, not a license to choose the denominator with the lower error. The paper-scope proxy relies on older proportional allocations that are not the Batty et al. completion algorithm adopted in 2025.

4.5 New-data robustness remains a separate result

The June 2026 public FWTW reconstruction remains informative. It produces 47.5% in 1963, 66.3% in 2007, a peak of 70.1% in 2009, and 66.2% in 2019. Its level rises from 157.2% to 325.6%. This later-vintage result follows the central trend, but it is not the same-input replication because it uses 21 rather than 23 intermediary definitions and later Financial Accounts revisions.

The defensible conclusion is therefore asymmetric. We have a close close old-vintage authors-data reproduction of Figure 1's level and a robust newer-data share pattern. We do not yet have the exact 2025 share denominator or completed annual matrices. No result here shows that the paper is wrong; making that claim would require the missing intended inputs and a reproducible failure of their stated identities.

Sources and reproducibility

The authoritative methodological source is Mian, Straub, and Sufi, *The Saving Glut of the Rich*, July 25, 2025 draft, Sections 2.1–2.2. The historical implementation source is the read-only February 2021 replication package, especially `MSS2021Febreplicationkit/code/build_fa_unveiling_data.do`. Project notation, code contracts, exact line crosswalks, and the current empirical boundary are documented in `docs/methods/leontief-unveiling.md`, `docs/data/figure1-data.md`, and `docs/reference/author-kit-map.md`. Numerical values in this report are generated by `Code/scripts/build_unveiling_teaching_assets.py`, `Code/scripts/build_figure1.py`, and `Code/scripts/build_figure1_authors_data.py`.